

The Essence of Entropy: the Degree of Energy Distribution Homogenization

-Entropy Increase: the Process of Energy Distribution Homogenization

-Multiplicative entropy:Section 3 of the Quantized Elastic Cosmic Model,
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Abstract:

How to understand the concept of Entropy?

Traditional understanding of entropy: disorder or randomness, which is a concept difficult to quantify. The multiplicative entropy proposed in this paper offers a computable and high-resolution definition of entropy within a quantized network, from a fundamentally calculable perspective.

Multiplicative entropy provides a precise way to characterize each step of state transitions in a topologically homeomorphic quantum space. In this framework, the space is composed of discrete units known as Space Elementary Quanta (SEQ), which form a dynamic quantum-elastic network. Each transformation of this network corresponds to a specific entropy value, calculated as the multiplicative product of the energy norms of all SEQ involved in that transformation. This means that every change in the spatial configuration—such as energy redistribution among SEQ—leads to a calculable increase in entropy. The total energy of the system remains constant, but the way it is distributed across SEQ changes, and this distribution becomes more uniform over time. The entropy reaches its maximum when all SEQ carry nearly equal energy, differing only by discrete Planck-scale units. This model ensures that each state transition is uniquely associated with an entropy value, making it possible to track the irreversible progression of time through the system's evolution. As energy spreads out and the distribution becomes more homogeneous, the multiplicative entropy increases step by step, reflecting the natural tendency of the system toward equilibrium. This approach not only defines entropy in a physically intuitive way but also links it directly to the microscopic dynamics of space itself, offering a foundation for understanding time's arrow in a quantum spacetime context.

3. Definition and analysis formula of entropy:

In this definition of entropy, the entropy value of a closed system at a given moment (i.e., during a specific state transition) is calculated as the multiplicative product of the energy norms of all SEQ involved in that transition(that moment's space transformation).

$$\text{Entropy value of closed system } S = \prod m_i, i \in N \quad (1)$$

$$\text{Energy of closed system} = \text{constant} = \sum m_i, i \in N \quad (2)$$

$$S_{\max} \leq m_i^n, \text{ When all } m_i \text{ are equal or differ only by Planck's constant } h$$

(Where m_i refers to the energy carried by the i th SEQ during a single transformation of the closed

system, where each energy state m_i is an integer multiple of Planck's constant h , $m_i=n_ih$, $n_i \in \mathbb{N}$)

Each SEQ carries energy m_i = the resonant kinetic energy of this SEQ+its spin kinetic energy+the elastic potential energy of the local space surrounding the SEQ, calculated based on the gradient of resonance frequency shift relative to the Planck frequency of the SEQ, as will be elaborated in Section 9.1 and Section13.

Under this setup, the total energy of the entire space can indeed be represented solely by m_i , without the need to separately calculate the elastic potential energy within the sub-Planckian components.

3.1 Energy transfer rules and triggering conditions:

Energy exchange occurs between adjacent SEQ (i,j) if and only if the following thermodynamic gradient exists: $m_i > m_j + h$, Energy transfer occurs only in discrete quanta of Planck's constant h , $m_i \rightarrow m_i - h$; $m_j \rightarrow m_j + h$ (Planck's constant: h)

3.2 Numerical Example: System States and Entropy Evolution

Table.1: Simplified Entropy Increase Demonstration

System State	SEQ Energy Distribution $E_{\text{total}} = \sum m_i = 12$	Entropy $S = \prod m_i$	Remarks
Initial non-equilibrium state	[3, 1, 5, 3]	45	-
Intermediate state	PathA[3, 1, 4, 4]; PathB[3, 2, 4, 3];	A48; B72;	-
Final state	PathA[3, 2, 3, 4]; PathB[3, 3, 3, 3];	A72; B81;	Due to adjacent energy transfer with minimal quanta h , this system cannot reach maximum entropy in case A

Note:The above analysis demonstrates that different entropy-increasing pathways exhibit distinct sequences of entropy variation.

3.3 Logarithmic Relation:

After logarithmic transformation, $\ln S$ aligns with the conventional Boltzmann entropy form, while the multiplicative formulation naturally suits discrete systems.

3.4 Proof of Spontaneous Entropy Increase

Spontaneity Theorem of entropy increase (Second Law of Thermodynamics):

For every possible energy transfer process, the total entropy change satisfies $\Delta S \geq 0$.

Proof Outline: Let the pre-transfer states be $m_i=a$, $m_j=b$ ($a > b+h$);

The post-transfer entropy ratio is:

$$S_{i1}/S_{i2} = (a-h)(b+h)/ab = 1 + h(a-b-h)/ab > 1 \text{ (Planck's constant } h)$$

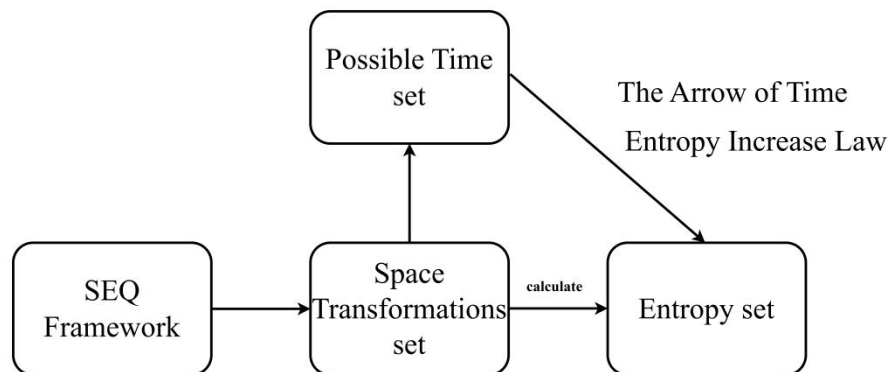
3.5 The definition conserves energy, has an entropy ceiling, ensures spontaneous increase, and logarithmically aligns with classical entropy.

Indeed, this model establishes a novel analytic quantum thermodynamic framework where coarse-graining automatically enforces the indistinguishability of quantized homogeneous

invariant network in physical properties while simultaneously characterizing the entropy-increasing process of energy homogenization. These properties collectively confirm the validity of this entropy definition. At this point, a clear multiple mapping can be established:

One space transformation→one entropy value→One possible moment

Space-Time-Entropy Mapping Diagram



Draw tool: draw.io [Computer software]. <https://github.com/jgraph/drawio>

Diagram.1: Space-Time-Entropy Mapping Diagram

3.6 The Spontaneous Entropy Increase is Causality.

The existence of isentropic transformations implies that even with the constraint of the least action principle further limiting the degrees of freedom in future spatial transformation paths, the potential spacetime evolution trajectories are not necessarily unique. Within this model's framework, multiple solutions may satisfy the least action condition, indicating that while causality adheres to Markovian properties, a certain dynamic freedom persists.

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3.7 Why Analytic Entropy is Adopted: Within this model's framework, physical states at Planck-time scales exhibit deterministic characteristics. Although current experimental conditions cannot directly measure them, traditional statistical entropy can be reduced to analytic expressions at this scale. This perspective not only reveals the microscopic essence of statistical quantities but also provides a new analytic foundation for quantum thermodynamics — unifying macroscopic statistical behaviors with deterministic dynamics at the Planck scale.

Multiplicative analytic entropy corresponds to each step of energy homogenization in a system. With every step toward a more uniform energy distribution, the multiplicative analytic entropy

increases, offering a clearer and more detailed characterization of the process compared to traditional statistical entropy.

Multiplicative Entropy vs. Traditional Statistical Entropy

Comparison Dimension	Multiplicative Entropy	Traditional Statistical Entropy
Process Explicitness	Explicitly records energy homogenization steps via product sequences (e.g., $\prod_i m_i$), preserving microstate transition details	Describes only macro-state differences via logarithmic state-count ($\ln \Omega$), erasing intermediate dynamics
Physical Intuitiveness	Entropy increase directly reflects irreversible energy redistribution; time asymmetry emerges from dynamics	Relies on probabilistic assumptions (e.g., molecular disorder) and requires ad hoc low-entropy past boundary
Process Resolution	Tracks Planck-timescale (t_p) energy transfers;	Limited to ensemble averages, incapable of resolving quantum fluctuations or short-timescale entropy production

3.8 Analysis of the Maximum Entropy Principle

In the SEQ model, the maximum entropy principle is manifested through the driving tendency of entropy increase: energy not only flows from a higher-energy SEQ to an adjacent lower-energy SEQ, but it also follows the path with the largest energy difference.

For example, consider an SEQ i , which is adjacent to two other SEQ j and k . The energies carried by nodes i , j , and k are A , B , and C respectively, where $A > B > C$. In this case, there are two possible energy transfer paths from SEQ i according to the principle of entropy increase: path $i \rightarrow j$ or path $i \rightarrow k$. We will analyze the entropy change for each path separately.

Under the constraint of energy conservation (i.e., $A + B + C = \text{constant}$), the entropy of the local system composed of these three SEQ before energy transfer is: $S = A \times B \times C$

Path $i \rightarrow j$: transferring energy to node j (which has relatively higher energy):

$$S' = (A - 1)(B + 1)C = (A - 1)(BC + C)$$

Path $i \rightarrow k$: transferring energy to node k (which has lower energy):

$$S'' = (A - 1)B(C + 1) = (A - 1)(BC + B)$$

Here, "1" represents one unit of Planck constant h . Since $B > C \Rightarrow BC + B > BC + C \Rightarrow S'' > S'$, the entropy increases more along the $i \rightarrow k$ path—that is, the path with the larger energy difference leads to a greater increase in entropy.

This deduction can be easily generalized to cases where the number of adjacent nodes is greater than two, so a detailed proof is omitted here.

However, it should be emphasized that the path with the maximum energy difference is not necessarily unique. Therefore, for states that have not yet occurred, the future still retains sufficient degrees of freedom — the evolution is not entirely deterministic.

3.9 In section 9.1.1, we will continue to discuss the decomposition of energy carried by SEQ, the relational expressions, and the supplementary description of the initial low-entropy state of the

universe.